

Cycling Fermi arc electrons with Weyl orbits

Cheng Zhang¹, Yi Zhang, Hai-Zhou Lu, X. C. Xie and Faxian Xiu²

Abstract | The Weyl orbit refers to a new type of cyclotron orbit recently proposed in Weyl semimetals. This is a surface–bulk hybrid orbit that can be modulated by thickness and external fields. Recently, the Weyl orbit has been used to realize the 3D quantum Hall effect, and subsequent rapid progress in experiments promise a new understanding of higher-dimension quantum Hall physics. In this Perspective, we review the theoretical background and the current experimental status of the Weyl orbit, present the extension of Weyl orbit physics in realizing the 3D quantum Hall effect and further discuss the opportunities in tuning topological quantum states and realizing device applications provided by the concept of the Weyl orbit.

The recent development of topological electronic systems introduces a new type of open Fermi surfaces — the Fermi arcs — as the surface states of the Dirac and Weyl semimetals^{1,2}. In contrast to the conventional open Fermi surfaces, the Fermi arc terminates at the projection of a bulk Weyl node pair inside the surface Brillouin zone. Naively, one may expect the absence of cyclotron orbits in Fermi arcs. However, Fermi arcs of opposite surfaces can couple together via the bulk chiral Landau levels in a slab geometry, and allow a new type of closed cyclotron orbit, termed the ‘Weyl orbit’, to develop³. This unique property of Fermi arcs results in the integration of the momentum-space cyclotron motion along a pair of spatially separated Fermi arcs and the position-space propagation between the two surfaces. The surface–bulk hybrid orbit is in stark contrast to the common cyclotron orbit along a simple closed loop in the momentum space as in typical Fermi pockets. This idea of the hybrid orbit reinterprets the current understanding of cyclotron orbits and gives rise to many intriguing physical consequences.

Unlike the typical cyclotron orbits around metallic Fermi surfaces, the propagation process of a Weyl orbit through the bulk along the chiral Landau levels parallel to the magnetic field contributes an additional thickness-related phase term in the Lifshitz–Onsager quantization rule^{3,4}. This unique property of the Weyl orbit

enables the modulation of Landau level energy through samples’ thickness and external magnetic fields^{3,4}. In addition, the Weyl orbit can support a distinctive 3D quantum Hall effect, since the inter-surface propagation is also dissipationless^{5,6}. Further, the Weyl-orbit-based quantum Hall effect supports a 3D distribution of edge states and a tunable quantum phase^{4,6}. These unique properties revive intense interests in quantum Hall systems in higher dimensions. The combination of topological boundary excitations and the quantum Hall effect in the Weyl orbit opens up a new frontier in the research of topological electronic materials.

In this Perspective, we present a brief overview of the theoretical background of the Fermi arcs and the Weyl orbit, followed by discussions on quantum transport evidence for the Weyl orbit. Then, we focus on the extension of Weyl orbit physics in realizing the 3D quantum Hall effect and review the current experimental progress. Next, we discuss the theoretical proposals and experimental advances in tuning topological quantum states and realizing device applications assisted by the Weyl orbit concept, and provide an outlook on future directions of the Weyl orbit.

Weyl semimetals and Fermi arcs

In 1928, Paul Dirac combined special relativity and quantum mechanics, and proposed a relativistic equation for the electron wave function⁷. The following

year, Hermann Weyl simplified the Dirac equation by considering a zero mass term $m = 0$ in an odd number of dimensions, to give a simplified form of the Hamiltonian,

$$H(\mathbf{k}) = \eta \mathbf{v} \boldsymbol{\sigma} \cdot \mathbf{k},$$

where η is the chirality. This form is known as the Weyl equation⁸ and it describes massless relativistic particles (Weyl fermions) with a definite chirality $\eta = \pm 1$. While such a particle remains to be discovered in high-energy physics, Weyl fermions have been realized as quasiparticle excitations in condensed matters^{1,9–15}.

In 3D solids, a band crossing point from two non-degenerate bands is stable against perturbations. In the vicinity of band crossing points, the dispersion is approximately linear, and the effective Hamiltonian takes the form of the Weyl equation, indicating the emergence of Weyl fermion excitations in solids. These band crossing points are the monopoles of Berry curvature $\boldsymbol{\Omega}$ (REF. 16). The chirality of a crossing point can be given by the Chern number (integral of the Berry curvature) of a Fermi surface that encloses the crossing point,

$$\eta = \frac{1}{2\pi} \int_{\Sigma} d\mathbf{S} \cdot \boldsymbol{\Omega} = \mp 1.$$

These band crossing points and the solid-state systems that host them are named Weyl nodes and Weyl semimetals, respectively^{1,17}. To ensure that a crystalline system has zero net chirality, the Weyl nodes always come in pairs of opposite chiralities. With a magnetic field along the z direction, the Weyl fermion forms Landau levels

$$\varepsilon_n = \eta v \text{sgn}(n) \sqrt{2|n|eB + k_z^2},$$

with n the Landau level index, v the Fermi velocity, e the elementary charge, B the magnetic field and k_z the Fermi wave vector along the z direction. Notably, there is a chiral (zeroth) Landau level, which disperses linearly at $\varepsilon = \eta v k_z$, with the sign of its Fermi velocity determined by the chirality η . We also note that Weyl semimetals have been proposed in even spatial dimensions¹⁸ by analogy with the point crossings between non-degenerate bands. However, these

scenarios are no longer covered by the original Weyl equation.

In systems with both time-reversal symmetry ($\mathcal{T}$) and spatial inversion symmetry ($\mathcal{P}$) preserved, the band dispersion $\varepsilon(\mathbf{k})$ satisfies Kramer's theorem $\varepsilon_+(\mathbf{k}) = \varepsilon_-(\mathbf{k})$, which means that the band is at least doubly degenerate. Thus, a Weyl semimetal must break either $\mathcal{T}$ or $\mathcal{P}$ symmetry. Correspondingly, material candidates for Weyl semimetals can be classified into two classes: $\mathcal{T}$ symmetry-breaking systems, such as pyrochlore iridates¹, HgCr_2Se_4 (REF.¹⁹), topological insulator multilayer¹⁷, YbMnBi_2 (REF.²⁰), half-Heusler compounds GdPtBi (REF.²¹) and magnetic Heusler compounds Co_2MnGa (REF.²²), $\text{Co}_3\text{Sn}_2\text{S}_2$ (REF.²³), Mn_3Sn (REFS^{24,25}), EuCd_2As_2 (REF.²⁶), and $\mathcal{P}$ symmetry-breaking systems, such as TaAs family^{9,10}, WTe_2 (REF.²⁷), MoTe_2 (REFS^{28,29}), TaIrTe_4 (REF.³⁰) and RhSi (REFS^{31,32}). Many of them have been confirmed in experiments^{11–15,20,24,25,33–41}. Note that, in semimetal systems with larger Chern numbers ($|C| > 1$), the energy band may no longer disperse linearly as described above; for example, $|C| = 2$ systems have quadratic dispersions^{19,31,32}. For the crossing of degenerate bands in $\mathcal{PT}$ symmetry-preserved systems, the Weyl nodes will annihilate in pairs, and a

gap opens upon perturbation. However, additional symmetries, such as the crystalline point-group symmetry, can stabilize the degeneracy and protect the nodes^{42–44}. In this case, the system becomes a Dirac semimetal, since the minimal model to depict the band crossing is a 4×4 Dirac Hamiltonian⁴⁵. Examples of Dirac semimetals include Na_3Bi (REFS^{43,46}), Cd_3As_2 (REFS^{44,47–49}), $\beta\text{-BiO}_2$ (REF.⁵⁰), Cu_3PdN (REFS^{51,52}) and PtTe_2 (REF.⁵³).

An important manifestation of the band topology is the emergence of boundary states, as widely observed in quantum Hall states and topological insulators. Considering a space enclosed by two 2D cross sections in the Brillouin zone of a 3D Weyl semimetal, the corresponding Chern number changes by ± 1 as the boundary moves across each Weyl node. Therefore, a gapless boundary channel appears between 2D cross sections enclosing non-zero net chirality (the internode interval of a Weyl node pair) and ends at the surface projection of the Weyl node pair. This boundary channel takes the form of a special arc-like open Fermi surface¹. The Fermi arcs (or Fermi surfaces in the case of finite Fermi energy) bridge the disconnected bulk Weyl nodes of opposite monopole charges, showing a chirality flow mediated by the Fermi arcs (FIG. 1a).

The observation of Fermi arcs can be used as an experimental diagnostic to identify the Weyl semimetal phase, as in the angle-resolved photoemission spectroscopy studies (ARPES) of TaAs (REFS^{11,12}) and MoTe_2 (REFS^{39–41}).

Dirac semimetals formed via the band inversion mechanism, also considered Z_2 Weyl semimetals⁵⁴, possess Dirac nodes generated in pairs, which can be viewed as two copies of superimposed Weyl nodes with opposite chirality. Thus, the Fermi arcs will also appear end to end in Dirac semimetals and form a fusiform shape in the surface Brillouin zone, which has been experimentally observed in Na_3Bi (REF.⁵⁵).

In this Perspective, we mainly focus on Fermi arc surface states and their interplay with the bulk chiral Landau levels. There are many other interesting properties of Weyl semimetals, such as the chiral anomaly^{56–67}, anomalous Hall/Nernst effect^{35,36,68–73}, nonlinear optical effect^{74,75}, magneto-optical response^{76–81}, axion electrodynamics^{58,82,83}, chiral zero sound^{84,85}, photogalvanic effect^{74,86–96} and Berry curvature dipoles^{97,98}, which are not covered here. Other topical reviews are available in the general field of topological semimetals^{2,99–116}.

Theory of Weyl orbits

A natural question that arises is how Fermi arc electrons respond to external magnetic fields. Semiclassically, electron wave packets in momentum space continuously evolve along the constant energy contour driven by the Lorentz force, either as a closed path or an open one across the adjacent Brillouin zones. On the contrary, the Fermi arcs terminate abruptly, which seems to violate the charge conservation and continuity conditions. However, once we take the bulk states into account, the Fermi arc ends where bulk states emerge, and the Fermi arc electrons can tunnel into the bulk.

Consider a Weyl semimetal with one pair of Weyl nodes in a slab geometry and a perpendicular magnetic field. There are two Fermi arcs on the top and bottom surfaces, respectively. The bulk states around the Weyl nodes form chiral Landau levels and disperse parallel to the magnetic field. Depending on the original Weyl node's chirality, the chiral Landau level acts as a one-way conveyor belt and transports the electrons from the top surface towards the bottom surface, or vice versa. After reaching the opposing surface, the electrons travel along the other Fermi arc before tunnelling back to the original surface via the counterpropagating chiral Landau level descending from the other Weyl node,

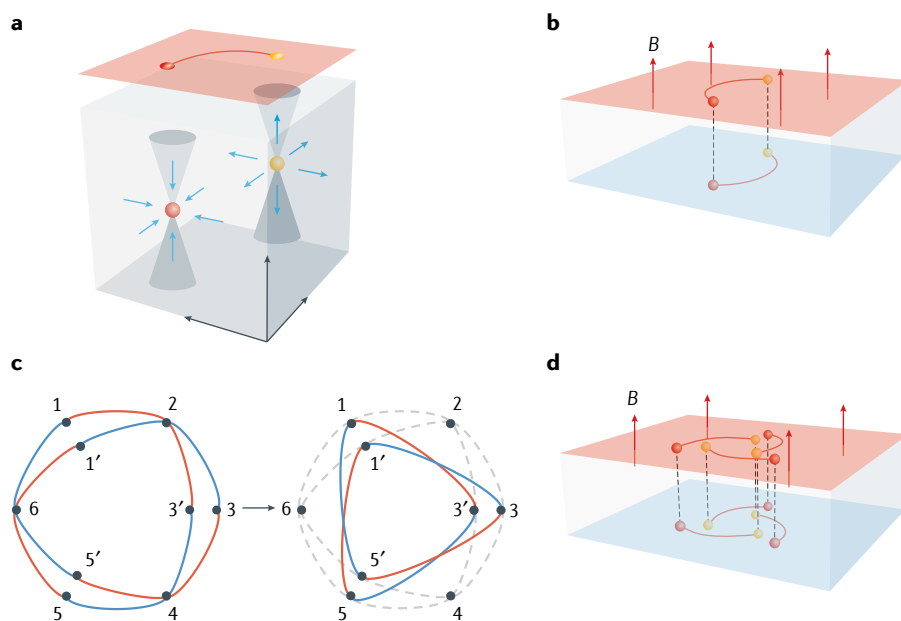


Fig. 1 | Schematics of Fermi arcs and Weyl orbits. **a** | The bulk and surface states of a minimum Weyl semimetal model with one pair of Weyl nodes. The Fermi arcs connect the surface projection of bulk Weyl nodes. **b** | Illustration of the Weyl orbit. The Fermi arcs located on opposite surfaces are linked by the bulk chiral Landau levels to accomplish a closed cyclotron loop. **c** | Generation of the trefoil-knot geometry from conventional Fermi arcs. Blue and orange arcs are located at the top and bottom surfaces, respectively. **d** | Three pairs of Fermi arcs criss-cross each other on the top and bottom surfaces of a slab and assemble a cyclotron orbit with a trefoil-knot geometry.

forming a closed, surface–bulk hybrid path (FIG. 1b).

This closed cyclotron path is the Weyl orbit³, and we can analyse its Landau level phenomena within the phase-space quantization framework⁴: the two Fermi arcs contribute a geometric phase proportional to the enclosed magnetic flux by the combined Fermi arcs; each of the two chiral Landau levels contributes a quantum phase $\oint \mathbf{p} d\mathbf{r} = k_{F\parallel} L$ proportional to its Fermi wave vector along the field direction ($k_{F\parallel}$), as well as the slab thickness (L); if the electrons' mean free path exceeds the scale of the Weyl orbit, for example, the slab thickness, the established phase coherence requires the overall phase accumulated across the Weyl orbit to satisfy the Bohr–Sommerfeld condition, which gives the Lifshitz–Onsager relation⁴:

$$\frac{\hbar S_k}{eB} + (k_{F1} - k_{F2})L = 2\pi(n + \gamma).$$

Here, n is the Landau level index, γ is a constant offset, L is the slab thickness, B is the magnetic field strength, e is the electron charge, $\hbar$ is the reduced Planck's constant, S_k is the momentum-space area enclosed by the combined Fermi arcs, and k_{F1} and k_{F2} are the Fermi wave vectors (which is the radius of the Fermi sphere in an isotropic case) of the two chiral Landau levels along the magnetic field, respectively. In comparison with conventional 2D cyclotron orbits, the contributions from the bulk propagations $(k_{F1} - k_{F2})L = (k_{W\parallel} + 2k_{F\parallel})L$ result in a thickness dependence for the Weyl orbit Landau levels. Here, $k_{W\parallel}$ is the separation between the pair of Weyl nodes and $k_{F\parallel} \approx v_F \varepsilon_F$ is the displacement of the Fermi wave vector from the corresponding Weyl nodes. Here, v_F and ε_F are the Fermi velocity and the Fermi energy, respectively.

Alternatively, we can derive the Weyl orbit Landau levels within the energy-time quantization framework³: taking the time that electrons spend in traversing the Fermi arcs $t_{\text{arc}} \approx 2k_0 / evB$ of length k_0 and the chiral Landau levels $t_{\text{bulk}} \approx 2L/v$, the energy of the n th Landau level should satisfy

$$\varepsilon_n \sim \frac{2\pi(n + \gamma)}{t_{\text{arc}} + t_{\text{bulk}}} = \frac{\pi v_F(n + \gamma)}{L + \frac{k_0}{eB}},$$

assuming an identical Fermi velocity v_F in the bulk and on the surface (Fermi arcs). With the caveat that ε_n is measured from the band bottom of the combined constant energy contour instead of the Weyl nodes⁴, which we commonly use to define the

Fermi energy ε_F , we obtain, once again, the quantization condition

$$\frac{1}{B_n} = \frac{e}{k_0} \left[\frac{\pi v}{\mu} (n + \gamma) - L \right].$$

These two theoretical approaches consistently show the quantization of the Weyl orbit and the thickness dependence of the resulting Landau levels. On the other hand, the period of quantum oscillations in the inverse of the magnetic field depends only on S_k and shows a 2D-like angle dependence as the magnetic field tilts, as only the surface Fermi arcs enclose magnetic flux.

A Weyl orbit can also consist of three Fermi arcs on the top surface and three Fermi arcs on the bottom surfaces that criss-cross each other through the bulk chiral Landau levels, realizing a trefoil-knot geometry¹⁷ (FIG. 1c,d). Such a Weyl orbit offers a new platform to explore the nontrivial knot theory in condensed matter and is topologically distinctive from the conventional cyclotron orbits in 2D or the simple Weyl orbit (FIG. 1b). In addition to the magnetic flux through the cyclotron orbit, the cyclotron orbit knot's self-threading property also introduces additional nontrivial flux along the orbit. Without the need for external magnetic fields, the phase of such Weyl orbits can also be induced by axial fields in systems with time-reversal symmetry¹⁸. It may be realized in strain-engineered electronic systems or other artificial systems, including photonic and acoustic crystals.

Quantum transport of Fermi arcs

As mentioned above, the Fermi arc surface states identified in ARPES experiments have provided key evidence for the Dirac and Weyl semimetal phases. The Fermi arc connects to the bulk state only at its terminal points, while the rest of the arc is gapped from the bulk states and fully localized on the surface. This unique, momentum-dependent surface–bulk connectivity was examined and observed in a scanning tunnelling microscope (STM) experiment¹⁹. The quasiparticle interference patterns obtained by STM identifies the scattering wave vectors of the surface electrons when encountering defects or impurities. STM experiments on the quasiparticle interference patterns of TaAs and NbP showed good agreement with theoretical calculations by taking account of the surface–bulk connectivity and the spin texture of the Fermi arcs^{120–122}.

Evidence of the Weyl orbit first emerged in magnetic quantum oscillations or Shubnikov–de Haas (SdH) oscillations in transport. Previous experiences in topological insulators suggest that one can enhance the surface conduction component by increasing the surface-to-volume ratio using nanostructures and thin films or tuning the Fermi level to the bulk gap by chemical doping or electrostatic gating^{123–127}. Owing to the semimetallic nature of bulk bands, the bulk density of states of topological semimetals can be suppressed as the Fermi level approaches the Weyl nodes. The first evidence of the Weyl orbit was achieved in thin films of Dirac semimetal Cd_3As_2 fabricated with the focused ion beam technique¹²⁸. A new frequency peak F_S (60–70 T) was observed in the fast Fourier transform spectra of quantum oscillations in addition to the original peak F_B (35–45 T) of the bulk Fermi surface in a series of Cd_3As_2 thin films with different thicknesses. By rotating the angle θ of magnetic fields in search of the Fermi surface anisotropy, F_S exhibits a clear 2D Fermi surface behaviour with $1/\cos(\theta)$ dependence (FIG. 2a), while F_B remains isotropic, as previously reported in Cd_3As_2 bulk crystals. As the film thickness increases above 2 μm , F_S gradually vanishes, likely owing to the suppression beyond the electron mean free path. In addition, a comparison of devices with triangular and rectangular cross-sectional geometries (FIG. 2b) provides further evidence for the Weyl orbit and its bulk propagation. In sharp contrast to a rectangular one, coherent quantum oscillations from the Weyl orbit are absent from a triangular device owing to the destructive interference between the immediate neighbours, as they contribute the thickness-dependent phases to the Lifshitz–Onsager relation. Hence, the surface oscillation F_S vanishes in the triangular device, whereas the bulk oscillation F_B persists.

The SdH oscillations of surface states in topological semimetals were also observed in nanostructures of Cd_3As_2 and NbAs grown by chemical vapour deposition^{129–132}, thin films of Cd_3As_2 grown by molecular beam epitaxy¹³³, exfoliated nanoribbons of WTe_2 (REF.¹³⁴) and polished micro-size crystals of TaAs (REF.¹³⁵). Compared with the simple band structure of Cd_3As_2 with two Dirac nodes, Weyl semimetals NbAs and TaAs host 12 pairs of Weyl nodes and complicated Fermi arc patterns⁹. It results in a high density of states at the surfaces and multiple sets of quantum oscillation frequencies. A systematic investigation of the Hall effect and surface state quantum

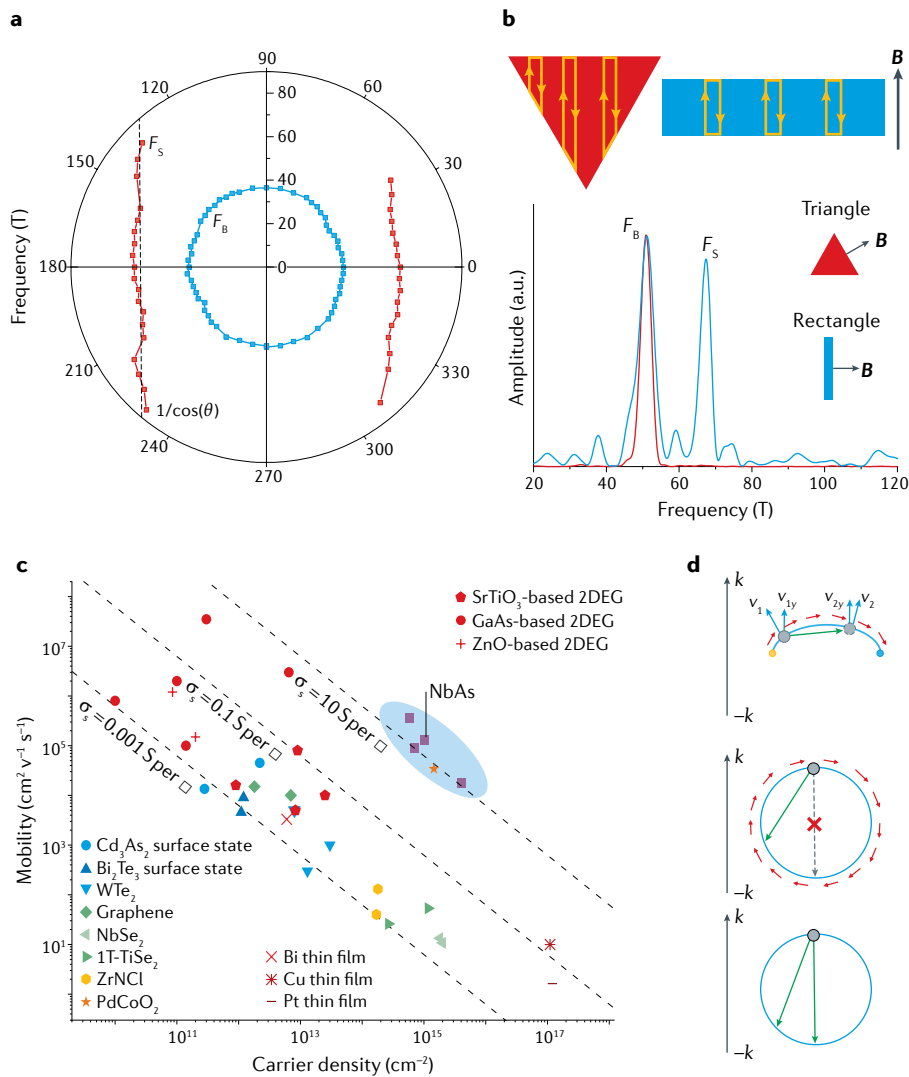


Fig. 2 | Quantum transport of the Weyl orbit in Dirac and Weyl semimetals. **a** | Polar plot of the angle dependence for the bulk F_B and the surface frequency F_S in a Cd_3As_2 thin film. F_B is nearly isotropic, while F_S shows an angle dependence of a 2D Fermi surface. **b** | Sketch of the Weyl orbits for rectangular and triangular geometries (top panel) and the frequency spectrum of the Shubnikov-de Haas oscillations in these two geometries (bottom panel). The measurement configurations are shown in the inset of the bottom panel. **c** | Comparison of the carrier density-mobility scaling among various 2D systems. **d** | The scattering processes in the momentum space for the Fermi arc (top), topological insulator surface state (middle) and conventional Fermi pocket without spin texture (bottom), respectively. The red, blue and green arrows indicate the spin texture, the velocities and the possible scattering vectors, respectively. 2DEG, two-dimensional electron gas. Panels **a** and **b** are reprinted from REF.¹²⁸, Springer Nature Limited. Panels **c** and **d** are adapted from REF.¹³¹, Springer Nature Limited.

oscillations showed that the surface states dominate the transport of NbAs nanobelts and host a record high sheet conductance up to 5–100 $\text{S}/\square$ (REF.¹³¹). For comparison, a typical sheet conductance of the most common 2D semiconductors, 2D metals and topological insulator surface states is two orders of magnitude smaller. A scaling plot of mobility versus carrier density among different 2D systems is presented in FIG. 2c. The surface states of NbAs can retain a high mobility of $1.5\sim 35 \times 10^4 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1}$, in spite of the high electron density on the

surface. Typically, mobility degrades as the carrier density increases, since a larger Fermi surface contour tends to enhance electron scattering. However, the Fermi arc surface states can overcome such a trade-off based on the following reasons. First, the intra-arc scattering is suppressed, since the velocities projected onto the direction of the current remain similar before and after scattering¹³⁶. Even if intra-arc scattering happens, it only slightly tilts the direction without strongly affecting the transport lifetime. FIGURE 2d illustrates the scattering

processes in the momentum space for a Fermi arc, a topological insulator surface state and a conventional Fermi pocket (the contour of Fermi energy in the Brillouin zone) without spin texture, respectively. Second, the inter-arc scattering is suppressed because the pair of Fermi arcs are spatially separated at opposite surfaces, while scattering between different pairs requires a large momentum transfer owing to the large displacement across the Brillouin zone between them for NbAs . Third, the bulk-arc scattering only happens near Weyl nodes, while the rest of the arcs are in the gapped regime of bulk states. For real topological semimetal materials without a strictly linear dispersion, the intra-arc scattering probability may become enhanced near the curved portion of each Fermi arc. The inter-arc scattering also depends on the momentum separation between separate Fermi arcs. In order to achieve high conductivity, Fermi arcs with a linear shape and large momentum separation are preferred. The low dissipation property of Fermi arcs offers new prospects towards electronic device applications of the topological semimetals.

In addition to the SdH oscillations, Aharonov–Bohm (AB) oscillations have also been observed in Cd_3As_2 nanowires^{137,138}, whose surface states form a cylindrical geometry. Periodic conductance oscillations appear as a function of the magnetic flux Φ enclosed by the electron paths owing to quantum interference. Observation of these interference-induced oscillations provides evidence for surface transport. When the magnetic field is applied along the longitudinal direction, the conductance in Cd_3As_2 nanowires oscillates with a periodicity of the flux quantum ($\Phi_0 = h/e$)¹³⁷. The AB oscillations also show a tunable phase shift when the Fermi level is modulated by the gate voltage¹³⁸. It results from the quantum-confinement-induced 1D surface subbands, whose energy $\epsilon(k_{\parallel}) = \pm \hbar v_F \sqrt{k_{\parallel}^2 + k_{\perp}^2}$, with $k_{\perp} = 2\pi(l + 0.5 - \Phi/\Phi_0)/C$, can be tuned by changing the magnetic flux and chemical potential^{137–139}. Here, 0.5 originates from the π Berry phase, l is the angular momentum quantum number and C is the circumference. By tuning the quantum phase, one can switch between a gapped mode and a gapless one in the AB oscillations. Such quasiballistic coherent transport observed in nanowires of Cd_3As_2 resembles the surface state transport in topological insulator nanoribbons¹³⁹ and demonstrates the topological nature of the surface states.

3D quantum Hall effect

The Weyl orbit can be used to induce a 3D quantum Hall effect, which is quantized across the entire thickness of the 3D sample. This is in contrast to previous proposals of 3D quantum Hall effect by dimensionality reduction, which rely on creating a stack of layers in the bulk and gapping out the interlayer dispersion, so that each layer contributes a quantized Hall response and adds up to a large sum of Hall conductance^{140–142}. The Weyl orbit version of the quantum Hall effect was firstly proposed using the Kubo formula⁵. To observe the conductance quantization, it is desirable to tune the Fermi energy to the Weyl nodes to suppress the bulk carriers and let the surface states dominate the transport.

The Weyl orbit induces Landau levels through surface–bulk hybridization.

The bulk propagation of the Weyl orbit is parallel to the magnetic field and incurs only an energy offset, μ_0 , to the quantized Landau levels for a constant thickness. The cyclotron of the Weyl orbit highly relies on the location of Weyl nodes and, therefore, shows the complex distribution of edge states. Let us consider a system with one pair of Weyl nodes separated along the k_y direction, in a magnetic field along the z direction (FIG. 3a). In position space, the skipping orbit along the x direction consists of the Fermi arcs on one surface, while the bulk propagation happens at the turning point of the skipping orbit, where the electrons meet an edge. For skipping orbits along other in-plane directions, such as the y direction, both the bulk chiral Landau levels and the side surfaces are involved. The corresponding edge state is, therefore,

extended along the z direction and intersects both the top and bottom surfaces. Hence, the Weyl orbit offers a new route towards the high-dimensional quantum Hall effect.

Experimentally, the quantum Hall effect has been observed in nanostructures and thin films of Dirac semimetal Cd_3As_2 (REFS^{129,132,133,143–149}). The current understanding of the quantum Hall effect in Cd_3As_2 can be divided into three scenarios for different experimental conditions, Weyl orbits, topological-insulator-type surface states or confined bulk states in the 2D limit. We discuss them separately in the following sections.

The Weyl orbit. Most of the nanostructures and thin films of Cd_3As_2 reported so far are crystallized in the (112) plane. Since the two pairs of overlapping Weyl nodes in Cd_3As_2

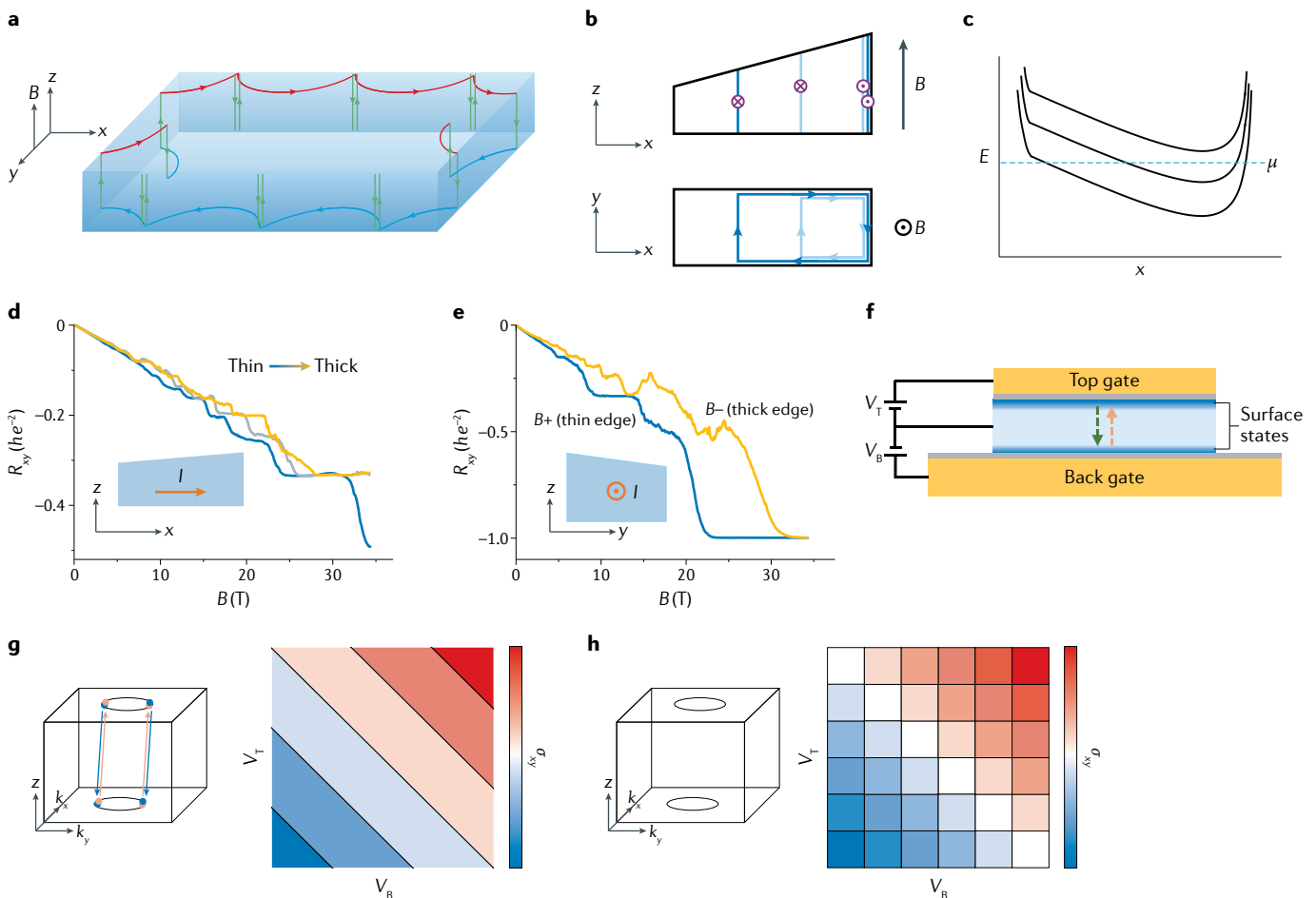


Fig. 3 | Quantum Hall effect of the Weyl orbit in Cd_3As_2 . **a** | Illustration of skipping edge states in a two-node Weyl semimetal. **b** | The x – z (top) and x – y (bottom) cross sections of a wedge geometry with varying thickness along the x direction. The blue lines correspond to quantum Hall chiral modes on the side walls and at certain critical thicknesses. **c** | The Landau level of the Weyl orbit with wedge geometry. **d,e** | Shifts of quantum Hall plateaux with thickness varying along the x (panel **d**) and y (panel **e**) directions, respectively. The insets are the measurement configurations. **f** | The schematic of a dual-gate device used to detect the inter-surface coupling

between the top and bottom surface states. **g** | Expected stripe regions (right), where different quantum Hall states appear in the 2D plot for surface states of two degenerated Weyl orbits (left). The filling factor difference between the adjacent patterns is 2. **h** | Expected chequerboard pattern (right), where different quantum Hall states appear in the 2D plot for the surface states of two uncoupled Dirac cones (left). The filling factor difference between the adjacent patterns is 1. Panel **a** is adapted with permission from REF.⁶, American Physical Society. Panels **b–e** are adapted from REF.¹³², Springer Nature Limited.

are along the [001] direction, each (112) surface hosts double Fermi arcs connecting end to end. Hence, there are two sets of Weyl orbits in Cd_3As_2 with the (112) surface. To elucidate the role of bulk propagation, a wedge geometry with thickness varying in one direction was adopted in experiments (FIG. 3b). According to the Lifshitz–Onsager relation of the Weyl orbit^{4,132}, the bulk-contributed quantum phase and Landau level will be thickness-dependent, resulting in a non-uniform filling factor ν along the wedge direction. Meanwhile, the thickness gradient needs to be small ($\alpha = \frac{dl}{dx} \ll O(0.1)$ estimated based on Cd_3As_2 nanoplate devices) to avoid destructive interference¹³².

The Landau level diagram suggests that the ‘edge’ states can arise in the central part of the surface without meeting a real edge (FIG. 3c). They are dispersive along the contour of equal thickness, localized along the gradient direction and extended along the thickness (FIG. 3b). When the wedge direction is parallel to the current I , a systematic shift of the quantum Hall plateau is observed at Hall electrode pairs at different thicknesses¹³² (FIG. 3d). By contrast, the quantum oscillations of bulk Fermi surface measured by in-plane magnetic fields and the low-field Hall effect are not affected by the thickness variation, excluding the non-uniform distribution of carrier density with the thickness.

When the wedge direction is perpendicular to the current I , the quantum Hall plateau positions shift as the magnetic field changes from positive to negative (FIG. 3e). This shift in position is due to the change of skipping orbit direction upon the reversal of the magnetic field. The quantum Hall chiral mode responsible for the charge transport switches to another edge, where the thickness is different¹³². Hence, the field-asymmetric behaviour of Hall resistance (FIG. 3e) is consistent with the Landau level shift along the longitudinal direction (FIG. 3d). Since the Weyl vector k_w in Cd_3As_2 breaks the mirror symmetry of Weyl orbits for the (44 $\bar{1}$) plane, asymmetric Landau level shifts are also observed as the magnetic field tilts away from the [112] direction towards opposite directions within the (1 $\bar{1}$ 0) plane¹³². It indicates that the Landau levels are also dependent on k_w . Such evidence together confirms the thickness-dependent Landau levels for the quantum Hall states of Cd_3As_2 , in agreement with the Weyl orbit scenario¹³². Importantly, the investigation of Landau level shifts with respect to the thickness on a single device not only coordinates the quantum Hall

plateaux with the oscillation peaks but also minimizes other impacts, such as surface contamination, edge directions or the bulk Fermi surfaces that differ from device to device.

Another important aspect of the Weyl orbit is the spatial distance between Fermi arcs at the two separated surfaces. By applying independent modulations such as gate voltage on two surfaces, one can extract the coupling effect between the surfaces and distinguish the Weyl orbit from other surface states (FIG. 3f). A recent study employed the dual-gate device configuration to tune the carrier density on the top and bottom surfaces of Cd_3As_2 thin films separately¹⁴⁹. For the Weyl orbit scenario, each gate modulates all the Weyl orbits simultaneously. Ignoring the degeneracy lifting of two Weyl orbits in Cd_3As_2 , the filling factor is expressed as $\nu = 2(\nu_1 + \frac{1}{2}) = \pm 1, \pm 3, \pm 5 \dots$ and corresponds to a stripe pattern extending along the constant sheet density line (FIG. 3g). By contrast, for other surface states as in topological insulators, each surface hosts a 2D electron gas independently and the quantum Hall filling factor ν is a simple summation of two surfaces (ν_1 and ν_2), with $\nu = (\nu_1 + \frac{1}{2}) + (\nu_2 + \frac{1}{2}) = 0, \pm 1, \pm 2 \dots$, which leads to a chequerboard quantum Hall plateau pattern in the 2D plot of top and bottom gates¹⁵⁰ (FIG. 3h). The experimental results in Cd_3As_2 thin films show a wide range of stripe patterns, in agreement with the Weyl orbit scenario¹⁴⁹. It is also consistent with the observation of one set of Landau levels in a previous experiment using a single-gate configuration¹⁴⁶. Further study of the evolution of filling factors in magnetic fields and gate voltages suggests complex Landau level crossings due to the field-induced splitting of two Weyl orbits¹⁴⁹.

Topological-insulator-type surface states.

There are two cases for the topological-insulator-type surface states in the quantum Hall regime of Cd_3As_2 . One is the (001)-oriented Cd_3As_2 grown by molecular beam epitaxy^{151,152}. The resultant (001) surface hosts topological-insulator-type surface states, since the surface projections of two Dirac nodes now overlap. It leads to a half-integer quantum Hall effect for each surface in the thin films. Interestingly, the quantum Hall state collapses into an insulating phase at large negative biases, possibly resulting from the strong electron correlation effect in high magnetic fields¹⁵².

The second case for the topological-insulator-type surface states considers the instability of surface Fermi arcs in

Dirac semimetals. Since Cd_3As_2 is a Dirac semimetal with two sets of overlapping Weyl node pairs, each surface hosts double Fermi arcs connecting end to end. If the crystal symmetry effect due to surface perturbation or magnetic fields is large enough, the double Fermi arc feature may deform into a Fermi pocket, similar to a Dirac cone in topological insulators^{3,153}. Then, the two inter-surface Weyl orbits in Cd_3As_2 will be replaced by two intra-surface cyclotron orbits without the bulk propagation process. It was argued that the transition from even to odd filling factors of the quantum Hall effect indicates a topological phase transition of the surface states, as the magnetic field tilts from the C_4 -symmetry-violated [112] to the C_4 -symmetry-preserved [001] direction¹⁴⁸. But the field-induced splitting mechanism observed in dual-gate Cd_3As_2 thin films also results in the even-to-odd change of filling factors¹⁴⁹. Recent calculations suggest that the surface Fermi arcs in Cd_3As_2 are quite robust to C_4 symmetry breaking and dimensional reduction¹⁵⁴. Therefore, it remains an open question as to whether the double Fermi arcs deform in high magnetic fields.

Confined bulk states in the 2D limit.

As the thickness of Cd_3As_2 approaches the 2D limit, the bulk band structure of Cd_3As_2 experiences a transition from a gapless Dirac semimetal to a gapped insulator⁴⁴. The gapped phase can be either topologically trivial or nontrivial, depending on the thickness regime⁴⁴. A study using a series of pulsed-laser-deposition-grown Cd_3As_2 thin films with a thickness ranging from 12 to 100 nm showed that the Fermi surface remains an isotropic 3D sphere at 100 nm but becomes 2D at 12–23 nm, with subband splitting and quantum Hall resistance¹⁴³. By extracting the energy–momentum relationship through quantum oscillation analysis, the system at the 2D limit matches the bulk state of a 2D topological insulating phase. Subsequent studies further utilized Zn doping to suppress the bulk carrier density¹⁴⁴ and successfully uncovered another quantum Hall regime corresponding to the Weyl orbit scenario^{145,149} in the 3D phase of Cd_3As_2 (~95 nm). In the quantum Hall states of the 3D phase, the bulk Fermi surface coexists with the surface quantum Hall effect, as revealed by quantum oscillations at in-plane magnetic fields, which is in agreement with the observation in Cd_3As_2 nanoplates.

As reviewed above, the intensive research on the topic of quantum Hall effect in Cd_3As_2 has gradually uncovered the complex phase diagram of Cd_3As_2 with

the modulation of thickness, magnetic fields and doping levels. To realize the Weyl-orbit-based 3D quantum Hall effect in Cd_3As_2 , a thickness range from 40 to 200 nm has been suggested to balance between bulk conduction and quantum confinement¹³². In addition, the 2D Cd_3As_2 at a smaller thickness is also intriguing owing to the tunable band inversion and quantum spin Hall phase via thickness variation and chemical doping. Further evidence of Fermi arc deformation in high magnetic fields is necessary towards the understanding of the symmetry breaking effect in Dirac semimetals. It is also of great interest to explore the quantum Hall effect in other topological semimetals, especially Weyl semimetal systems with more robust Fermi arc states.

Emergent quantum phenomena

As the first cyclotron orbit that combines the momentum-space cyclotron and the position-space propagation, the Weyl orbit physics offers possibilities towards realizing many emergent quantum phenomena and device applications. FIGURE 4a,b presents two device designs in which the Weyl orbit

strongly modifies the transport properties of the system. A magnetic field is applied along the z direction and induces Weyl orbits across the top and bottom surfaces. The first one (FIG. 4a, top panel) is in d.c. configuration. A d.c. current is injected from two adjacent electrodes from the top surface. Owing to the cyclotron loop of the Weyl orbit, an inter-surface current path is constructed when a magnetic field is applied. This current path violates Ohm's law and leads to a nonlocal current in the opposite direction between the bottom electrodes¹⁵⁵. The second one (FIG. 4a, bottom panel) is in a.c. configuration with electromagnetic microwave radiation propagating from top to bottom. The thickness of the sample is set to be much larger than the penetration depth of the microwave. The system is found to reject the transmission of electromagnetic microwave at most frequencies except for $\omega = n\omega_0$, with ω_0 the cyclotron frequency of the Weyl orbit and $n = 1, 2, 3, \dots$ (REF.¹⁵⁵). In other words, the Weyl orbit realizes a narrow bandpass filter by resonant transmission with magnetic field tunability.

The Weyl orbit can also help to establish a strong nonlocal correlation effect between

two potential wells residing near the two surfaces, which may enable remote quantum control effect in mesoscopic devices¹⁵⁶. The intriguing quantum Hall effect based on the Weyl orbit provides a new concept to realize quantized transport in 3D systems. Its edge states host a 3D distribution as discussed above (FIG. 3a). A recent theoretical study further finds that the location of the skipping edge state can be tuned to opposite sides by tilting the magnetic field towards certain directions⁶ (FIG. 4b). With the skipping edge states switching sides, there is a sudden sign change of the Hall conductivity for $\mathcal{T}$ symmetry-breaking Weyl systems. The application of in-plane magnetic fields $\mathbf{B}$ and pseudo-magnetic fields $\mathbf{B}_s$ (effective magnetic fields caused by lattice deformation) can lead to a dispersion of chiral Landau levels (along the direction of $\mathbf{B}$) and Fermi arcs (along the surface of $\mathbf{B}_s$), respectively^{157–159}. By engineering the strength of $\mathbf{B}$ and $\mathbf{B}_s$, one can obtain a range of Weyl orbit geometries, such as rectangle, trapezoid and butterfly shapes^{160,161} (FIG. 4c). Similar twisted Fermi surfaces with a butterfly shape can also be formed in Weyl semimetal thin films with an

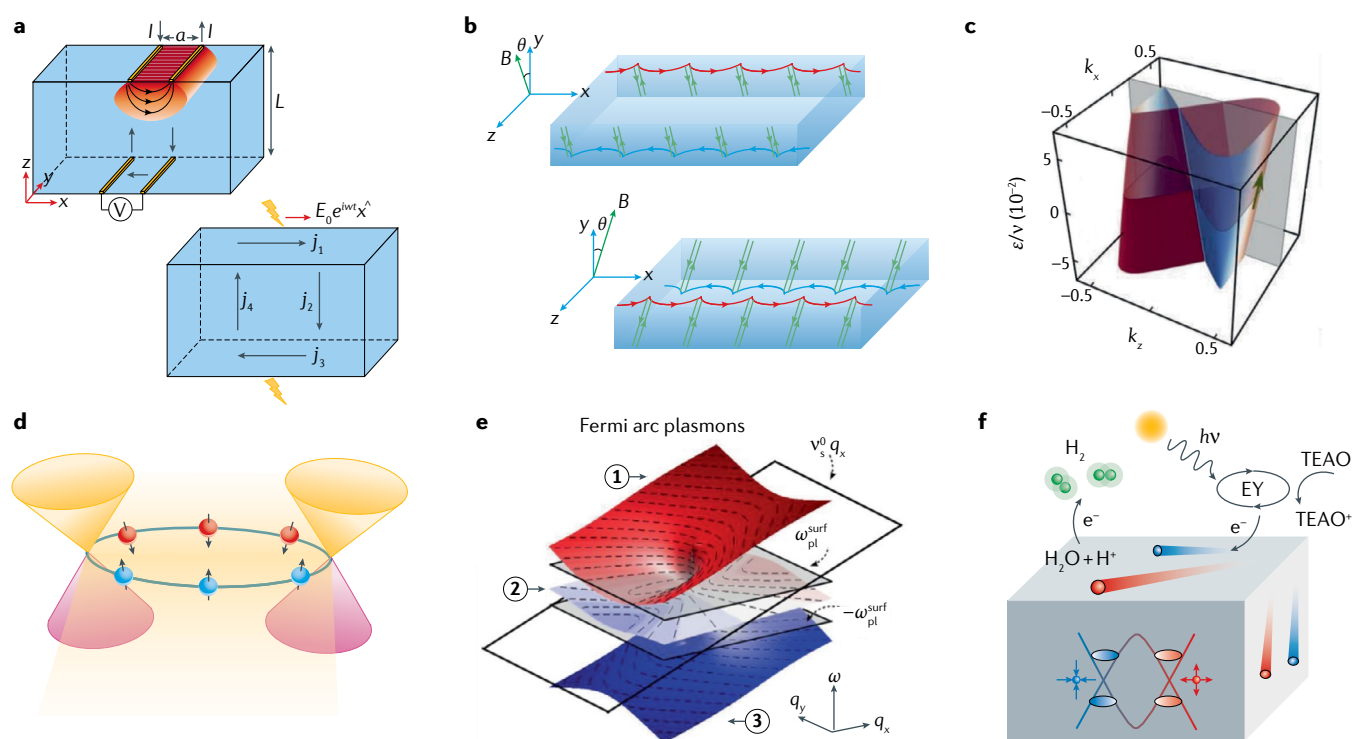


Fig. 4 | Emergent quantum phenomena and device applications based on the Weyl orbit. **a** | The d.c. (top) and a.c. (bottom) configurations of non-local conductivity assisted by the Weyl orbit. **b** | The tilt of magnetic fields in the y - z plane switches the distributions of skipping edge state in a two-node Weyl semimetal model. **c** | The dispersion of twisted Fermi surfaces induced by the tuning of in-plane magnetic fields and pseudo-magnetic fields. **d** | The Cooper pairs formed between the double Fermi arcs in Cd_3As_2 .

e | The dispersion of Fermi arc plasmons. **f** | Schematic diagram of Weyl semimetals for sensitive hydrogen evolution catalysts. Panel **a** is reprinted with permission from REF.¹⁵⁵, American Physical Society. Panel **b** is adapted with permission from REF.⁶, American Physical Society. Panel **c** is reprinted with permission from REF.¹⁶⁰, American Physical Society. Panel **d** is adapted from REF.¹⁶⁷, CC BY 4.0. Panel **e** is reprinted with permission from REF.¹⁷⁹, American Physical Society. Panel **f** is reprinted with permission from REF.¹⁸⁰, Wiley.

in-plane magnetization and broken spatial inversion symmetry¹⁶². Together with the trefoil-knot Weyl orbit¹¹⁷ mentioned earlier, the spatial separation property of the Weyl orbit offers a route towards the realization of twisted Fermi surfaces, which lead to self-intersecting cyclotrons with opposite AB phases and counterpropagating edge states with unusual magnetotransport behaviours¹⁶².

Recent studies on Weyl orbits also indicate the possibility of realizing the quantum-limit Hall effect (the filling factor $\nu < 1$) in topological semimetals with relatively small magnetic fields¹⁶³. Unlike a conventional Fermi pocket, which is either electron-type or hole-type, depending on the sign of the Fermi velocity, the surface–bulk dichotomy of the Weyl orbit makes it natural to have opposite carrier types in different sections along a single closed contour, for example, hole-type surface Fermi arcs yet electron-type bulk states around the Weyl nodes, or vice versa. The cancellation between charges of different types can suppress the Landau level index $\nu = (n_e - n_h)h/eB$, even in the presence of large carrier density residues n_e and n_h , especially when $n_e \sim n_h$. Such a mechanism in the Weyl orbit may help relax the demanding requirements of large magnetic fields and small carrier densities for the quantum limit and broadly promote its experimental realizations and applications.

Another interesting direction of employing the Weyl orbit physics is the combination of the Weyl orbit with other effects, such as superconductivity, light–matter interaction and symmetry breaking. A series of experiments have been devoted to study the superconductivity proximity effect and intrinsic superconductivity induced by external modulations of topological semimetals^{164–178}. Evidence of surface superconductivity has been reported in Cd_3As_2 nanoplates and nanowires with Nb contacts through the evolution of the supercurrent quantum interference pattern^{167,172,173}, 4π -periodic supercurrent¹⁷⁰ and a Fano interference feature¹⁷¹. The pairing mechanism for the double Fermi arcs in Cd_3As_2 systems is illustrated in FIG. 4d. Since the Fermi arcs originate from edge states of 2D topological insulators embedded in the topological semimetal, the surface superconductivity in Cd_3As_2 can be regarded as a higher-dimensional analogue of edge superconductivity in 2D quantum spin Hall insulators and provide a new promising platform for realizing Majorana flat bands¹⁶⁷. Generalizing the Weyl orbit

physics to the Cooper pairing may lead to an inter-surface pairing of Fermi arcs with time-reversed momenta and spins, which is another manifestation of the nonlocal correlation effect in the Weyl orbit. The collective dynamics of electrons in Fermi arcs is also distinct from conventional metals. The collective modes of Fermi arc electrons can hybridize with conventional surface plasmons owing to the free carriers in the bulk. This hybridization leads to a plasmon dispersion of three chiral branches with hyperbolic frequency contours over a wide bandwidth, which can be probed by scanning near-field optical microscopy¹⁷⁹ (FIG. 4e). The hyperbolic character of Fermi arc plasmons can be useful in planar photonics. An unexpected application of the Fermi arc surface states arises in the field of chemical catalysts for hydrogen and oxygen evolution^{180–184} (FIG. 4f). Here, the Fermi arcs serve as a stable supply of itinerant electrons with high conductivity and high mobility properties for the catalysis process. It will also be of interest to explore the possibility of using Weyl semimetal thin films as an interface membrane for high-efficiency charge exchange in electrochemistry via the formation of Weyl orbits under real or pseudo-magnetic fields.

Outlook

The Weyl orbit not only renovates the concept of cyclotron orbits but also provides opportunities for exploring physical phenomena and device applications that are not otherwise accessible. We conclude this Perspective by discussing future research prospects enabled by the development of the Weyl orbit physics.

Quantum Hall effect in Cd_3As_2 . Research on the quantum Hall effect in Cd_3As_2 has moved forwards rapidly in recent years, but there are still many open questions in the field. In magnetic fields, the double Fermi arcs in Cd_3As_2 can be affected by symmetry breaking perturbations and deformed to conventional Fermi pockets. Further investigations are required to clarify the nature of the quantum Hall effect of Cd_3As_2 in high magnetic fields. Despite a finite bulk Fermi surface, the Hall conductance of 3D films of Cd_3As_2 can reach a good quantization value. It remains unclear how the bulk carriers are depleted in perpendicular magnetic fields. The complex phase diagram of Cd_3As_2 includes the intrinsic Dirac semimetal phase, symmetry-breaking-induced topological insulator phase and the dimension-reduction-induced quantum spin Hall insulator phase. Each

of them corresponds to a scenario of the quantum Hall effect as discussed above and all of them are very interesting in their own way. Cd_3As_2 is, therefore, a fruitful platform for exploring the combination of topological boundary excitations and the quantum Hall effect.

New systems for the Weyl orbit. Despite a wide range of topological semimetals discovered so far, only Cd_3As_2 , NbAs, WTe_2 and TaAs have been reported to host transport evidence for the Weyl orbit^{128,129,131,132,134,135,145,146,152}. These are either Dirac semimetals or $\mathcal{P}$ symmetry-breaking Weyl semimetals. The $\mathcal{T}$ symmetry-breaking Weyl semimetal systems host a net chirality flow in the Brillouin zone and can result in novel phenomena absent in $\mathcal{T}$ symmetry-preserving systems, such as the field-tunable edge state location and quantum Hall conductance contributed by the bulk chiral Landau levels⁶. Hence, it is of great interest to search for Weyl orbits and quantum Hall effect in $\mathcal{T}$ symmetry-breaking Weyl semimetals, as well as other topological semimetal systems, including node-line semimetals¹⁸⁵ and chiral semimetals. Pseudo-electromagnetic fields can be generated by strain, acoustic propagation and inhomogeneous magnetization, which enables the formation of Weyl orbits at zero magnetic fields¹⁶¹. The co-propagating chiral Landau levels induced by the pseudo-electromagnetic fields will have an impact on the resultant Weyl orbits^{118,161}.

Nonlocal correlation of Fermi arcs. The Weyl orbit bridges the two spatially separated Fermi arcs through the bulk chiral Landau levels. It is stimulating to investigate how conventional short-range interactions, such as the superconducting pairing, or the excitonic pairing evolve in the framework of the Weyl orbit with nonlocal correlation properties. From the perspective of device applications, the nonlocal correlation results in the magical transmission of electrical current and electromagnetic waves¹⁵⁵, and builds a long-range coupling between potential wells located at the two surfaces¹⁵⁶. It can be seen as using the Weyl orbit to construct a wormhole across a mesoscopic scale to exchange quasiparticles and information.

Light–matter interaction in Fermi arcs.

The unusual Fermi arc surface states also promise many novel light–matter interaction effects. The optical conductivity of Fermi arcs shows a strong anisotropy owing to

the arc-type configuration¹⁸⁶. Fermi arc photocurrents of different mechanisms are predicted and observed in topological semimetals^{90,187}. The formation of the Weyl orbit in magnetic fields will also influence optical spectroscopy and photocurrent. Compared with transport, the optical approaches enable access to both occupied and unoccupied states with a high time domain resolution, which can illuminate the helicoid dispersion and novel collective dynamics of Fermi arcs.

By reviewing the current progress of the theoretical and experimental work, we hope to provide a general framework of the Weyl orbit research. The ongoing investigations of the Weyl orbit in topological semimetals indicate a nascent yet fertile field. Going forwards, we expect emergent physics and utilities as one combines the concept of the Weyl orbit with other exotic effects and develops functional devices based on the Weyl orbit physics.

Cheng Zhang^{1,2,3}, Yi Zhang⁴, Hai-Zhou Lu^{5,6}, X. C. Xie⁴ and Faxian Xiu^{1,2,3,7,8,9}

¹State Key Laboratory of Surface Physics and Department of Physics, Fudan University, Shanghai, China.

²Institute for Nanoelectronic Devices and Quantum Computing, Fudan University, Shanghai, China.

³Zhangjiang Fudan International Innovation Center, Fudan University, Shanghai, China.

⁴International Center for Quantum Materials, School of Physics, Peking University, Beijing, China.

⁵Institute for Quantum Science and Engineering and Department of Physics, South University of Science and Technology of China, Shenzhen, China.

⁶Shenzhen Key Laboratory of Quantum Science and Engineering, Shenzhen, China.

⁷Mat-X Institute of Applied Materials, Fudan University, Shanghai, China.

⁸Shanghai Research Center for Quantum Sciences, Shanghai, China.

⁹e-mail: Faxian@fudan.edu.cn

<https://doi.org/10.1038/s42254-021-00344-z>

Published online 15 July 2021

- Wan, X., Turner, A. M., Vishwanath, A. & Savrasov, S. Y. Topological semimetal and Fermi arc surface states in the electronic structure of pyrochlore iridates. *Phys. Rev. B* **83**, 205101 (2011).
Theoretical proposition of Weyl semimetal candidate and Fermi arc surface states.
- Armitage, N. P., Mele, E. J. & Vishwanath, A. Weyl and Dirac semimetals in three-dimensional solids. *Rev. Mod. Phys.* **90**, 015001 (2018).
This is a comprehensive review of Weyl and Dirac semimetals.
- Potter, A. C., Kimchi, I. & Vishwanath, A. Quantum oscillations from surface Fermi arcs in Weyl and Dirac semimetals. *Nat. Commun.* **5**, 5161 (2014).
This paper proposed the concept of the Weyl orbit.
- Zhang, Y., Bulmash, D., Hosur, P., Potter, A. C. & Vishwanath, A. Quantum oscillations from generic surface Fermi arcs and bulk chiral modes in Weyl semimetals. *Sci. Rep.* **6**, 23741 (2016).
- Wang, C. M., Sun, H.-P., Lu, H.-Z. & Xie, X. C. 3D quantum Hall effect of Fermi arcs in topological semimetals. *Phys. Rev. Lett.* **119**, 136806 (2017).
This paper proposed the 3D quantum Hall effect based on the Weyl orbit.

- Li, H., Liu, H., Jiang, H. & Xie, X. C. 3D quantum Hall effect and a global picture of edge states in Weyl semimetals. *Phys. Rev. Lett.* **125**, 036602 (2020).
- Dirac, P. A. M. The quantum theory of the electron. *Proc. R. Soc. Lond. A Math. Phys. Eng. Sci.* **117**, 610–624 (1928).
- Weyl, H. Elektron und Gravitation. I. *Z. Phys.* **56**, 330–352 (1929).
- Weng, H., Fang, C., Fang, Z., Bernevig, B. A. & Dai, X. Weyl semimetal phase in noncentrosymmetric transition-metal monophosphides. *Phys. Rev. X* **5**, 011029 (2015).
- Huang, S.-M. et al. A Weyl Fermion semimetal with surface Fermi arcs in the transition metal mononitride TaAs class. *Nat. Commun.* **6**, 7373 (2015).
Weng et al. (Phys. Rev. X, 2015) and Huang et al. (Nat. Commun., 2015) proposed the inversion-symmetry-breaking Weyl semimetal TaAs class.
- Xu, S.-Y. et al. Discovery of a Weyl fermion semimetal and topological Fermi arcs. *Science* **349**, 613–617 (2015).
- Lv, B. Q. et al. Experimental discovery of Weyl semimetal TaAs. *Phys. Rev. X* **5**, 031013 (2015).
Xu et al. (Science, 2015a) and Lv et al. (Phys. Rev. X, 2015) report the experimental evidence of Fermi arcs in Weyl semimetals.
- Xu, S.-Y. et al. Discovery of a Weyl fermion state with Fermi arcs in niobium arsenide. *Nat. Phys.* **11**, 748–754 (2015).
- Yang, L. X. et al. Weyl semimetal phase in the non-centrosymmetric compound TaAs. *Nat. Phys.* **11**, 728–732 (2015).
- Lv, B. Q. et al. Observation of Weyl nodes in TaAs. *Nat. Phys.* **11**, 724–727 (2015).
- Fang, Z. et al. The anomalous Hall effect and magnetic monopoles in momentum space. *Science* **302**, 92–95 (2003).
- Burkov, A. A. & Balents, L. Weyl semimetal in a topological insulator multilayer. *Phys. Rev. Lett.* **107**, 127205 (2011).
- Ahn, J. & Yang, B.-J. Unconventional topological phase transition in two-dimensional systems with space-time inversion symmetry. *Phys. Rev. Lett.* **118**, 156401 (2017).
- Xu, G., Weng, H., Wang, Z., Dai, X. & Fang, Z. Chern semimetal and the quantized anomalous Hall effect in HgCr₂Se₄. *Phys. Rev. Lett.* **107**, 186806 (2011).
- Borisenko, S. et al. Time-reversal symmetry breaking type-II Weyl state in YbMnBi₂. *Nat. Commun.* **10**, 3424 (2019).
- Hirschberger, M. et al. The chiral anomaly and thermopower of Weyl fermions in the half-Heusler GdPtBi. *Nat. Mater.* **15**, 1161–1165 (2016).
- Chang, G. et al. Topological Hopf and chain link semimetal states and their application to Co₂MnGa. *Phys. Rev. Lett.* **119**, 156401 (2017).
- Xu, Q. et al. Topological surface Fermi arcs in the magnetic Weyl semimetal Co₃Sn₂S₂. *Phys. Rev. B* **97**, 235416 (2018).
- Nakatsuji, S., Kiyohara, N. & Higo, T. Large anomalous Hall effect in a non-linear antiferromagnet at room temperature. *Nature* **527**, 212–215 (2015).
- Kuroda, K. et al. Evidence for magnetic Weyl fermions in a correlated metal. *Nat. Mater.* **16**, 1090–1095 (2017).
- Ma, J.-Z. et al. Spin fluctuation induced Weyl semimetal state in the paramagnetic phase of EuCd₂As₂. *Sci. Adv.* **5**, eaaw4718 (2019).
- Soluyanov, A. A. et al. Type-II Weyl semimetals. *Nature* **527**, 495–498 (2015).
- Sun, Y., Wu, S.-C., Ali, M. N., Felser, C. & Yan, B. Prediction of Weyl semimetal in orthorhombic MoTe₂. *Phys. Rev. B* **92**, 161107 (2015).
- Wang, Z. et al. MoTe₂: a type-II Weyl topological metal. *Phys. Rev. Lett.* **117**, 056805 (2016).
- Koepnick, K. et al. TaIrTe₄: a ternary type-II Weyl semimetal. *Phys. Rev. B* **93**, 201101 (2016).
- Bradlyn, B. et al. Beyond Dirac and Weyl fermions: unconventional quasiparticles in conventional crystals. *Science* **353**, aaf5037 (2016).
- Chang, G. et al. Unconventional chiral fermions and large topological Fermi arcs in RhSi. *Phys. Rev. Lett.* **119**, 206401 (2017).
- Belopolski, I. et al. Discovery of topological Weyl fermion lines and drumhead surface states in a room temperature magnet. *Science* **365**, 1278–1281 (2019).
- Belopolski, I. et al. Signatures of a time-reversal symmetric Weyl semimetal with only four Weyl points. *Nat. Commun.* **8**, 942 (2017).
- Liu, E. et al. Giant anomalous Hall effect in a ferromagnetic kagome-lattice semimetal. *Nat. Phys.* **14**, 1125–1131 (2018).
- Wang, Q. et al. Large intrinsic anomalous Hall effect in half-metallic ferromagnet Co₂Sn₂S₇ with magnetic Weyl fermions. *Nat. Commun.* **9**, 3681 (2018).
- Liu, D. F. et al. Magnetic Weyl semimetal phase in a Kagomé crystal. *Science* **365**, 1282–1285 (2019).
- Morali, N. et al. Fermi-arc diversity on surface terminations of the magnetic Weyl semimetal Co₃Sn₂S₇. *Science* **365**, 1286–1291 (2019).
- Huang, L. et al. Spectroscopic evidence for a type II Weyl semimetallic state in MoTe₂. *Nat. Mater.* **15**, 1155–1160 (2016).
- Deng, K. et al. Experimental observation of topological Fermi arcs in type-II Weyl semimetal MoTe₂. *Nat. Phys.* **12**, 1105–1110 (2016).
- Jiang, J. et al. Signature of type-II Weyl semimetal phase in MoTe₂. *Nat. Commun.* **8**, 13973 (2017).
- Fang, C., Gilbert, M. J., Dai, X. & Bernevig, B. A. Multi-Weyl topological semimetals stabilized by point group symmetry. *Phys. Rev. Lett.* **108**, 266802 (2012).
- Wang, Z. et al. Dirac semimetal and topological phase transitions in A₃Bi (A = Na, K, Rb). *Phys. Rev. B* **85**, 195320 (2012).
- Wang, Z., Weng, H., Wu, Q., Dai, X. & Fang, Z. Three-dimensional Dirac semimetal and quantum transport in Cd₃As₂. *Phys. Rev. B* **88**, 125427 (2013).
The above two papers predict the most widely studied Dirac semimetal candidates, Na₃Bi and Cd₃As₂.
- Yang, B. J. & Nagaosa, N. Classification of stable three-dimensional Dirac semimetals with nontrivial topology. *Nat. Commun.* **5**, 4898 (2014).
- Liu, Z. K. et al. Discovery of a three-dimensional topological Dirac semimetal, Na₃Bi. *Science* **343**, 864–867 (2014).
An ARPES study of Dirac semimetals.
- Liu, Z. K. et al. A stable three-dimensional topological Dirac semimetal Cd₃As₂. *Nat. Mater.* **13**, 677–681 (2014).
- Neupane, M. et al. Observation of a three-dimensional topological Dirac semimetal phase in high-mobility Cd₃As₂. *Nat. Commun.* **5**, 3786 (2014).
- Borisenko, S. et al. Experimental realization of a three-dimensional Dirac semimetal. *Phys. Rev. Lett.* **113**, 027603 (2014).
- Young, S. M. et al. Dirac semimetal in three dimensions. *Phys. Rev. Lett.* **108**, 140405 (2012).
- Yu, R., Weng, H., Fang, Z., Dai, X. & Hu, X. Topological node-line semimetal and Dirac semimetal state in antiperovskite Cu₃PdN. *Phys. Rev. Lett.* **115**, 036807 (2015).
- Kim, Y., Wieder, B. J., Kane, C. L. & Rappe, A. M. Dirac line nodes in inversion-symmetric crystals. *Phys. Rev. Lett.* **115**, 036806 (2015).
- Yan, M. et al. Lorentz-violating type-II Dirac fermions in transition metal dichalcogenide PtTe₂. *Nat. Commun.* **8**, 257 (2017).
- Morimoto, T. & Furusaki, A. Weyl and Dirac semimetals with Z₂ topological charge. *Phys. Rev. B* **89**, 235127 (2014).
- Xu, S.-Y. et al. Observation of Fermi arc surface states in a topological metal. *Science* **347**, 294–298 (2015).
- Nielsen, H. & Ninomiya, M. The Adler-Bell-Jackiw anomaly and Weyl fermions in a crystal. *Phys. Lett. B* **130**, 389–396 (1983).
- Zyuzin, A. A. & Burkov, A. A. Topological response in Weyl semimetals and the chiral anomaly. *Phys. Rev. B* **86**, 115133 (2012).
- Wang, Z. & Zhang, S.-C. Chiral anomaly, charge density waves, and axion strings from Weyl semimetals. *Phys. Rev. B* **87**, 161107 (2013).
- Son, D. T. & Spivak, B. Z. Chiral anomaly and classical negative magnetoresistance of Weyl metals. *Phys. Rev. B* **88**, 4807–4813 (2012).
- Xiong, J. et al. Evidence for the chiral anomaly in the Dirac semimetal Na₃Bi. *Science* **350**, 413–416 (2015).
- Li, Q. et al. Chiral magnetic effect in ZrTe₅. *Nat. Phys.* **12**, 550–554 (2016).
- Huang, X. et al. Observation of the chiral-anomaly-induced negative magnetoresistance in 3D Weyl semimetal TaAs. *Phys. Rev. X* **5**, 031023 (2015).
- Zhang, C.-L. et al. Signatures of the Adler-Bell-Jackiw chiral anomaly in a Weyl fermion semimetal. *Nat. Commun.* **7**, 10735 (2016).
- Zhang, C. et al. Room-temperature chiral charge pumping in Dirac semimetals. *Nat. Commun.* **8**, 13741 (2017).
- Li, H. et al. Negative magnetoresistance in Dirac semimetal Cd₃As₂. *Nat. Commun.* **7**, 10301 (2016).

66. Li, C. Z. et al. Giant negative magnetoresistance induced by the chiral anomaly in individual Cd_3As_2 nanowires. *Nat. Commun.* **6**, 10137 (2015).
67. Yuan, X. et al. The discovery of dynamic chiral anomaly in a Weyl semimetal NbAs. *Nat. Commun.* **11**, 1259 (2020).
68. Yang, K.-Y., Lu, Y.-M. & Ran, Y. Quantum Hall effects in a Weyl semimetal: possible application in pyrochlore iridates. *Phys. Rev. B* **84**, 075129 (2011).
69. Burkov, A. A. Anomalous Hall effect in Weyl metals. *Phys. Rev. Lett.* **113**, 187202 (2014).
70. Liang, T. et al. Anomalous Hall effect in ZrTe_5 . *Nat. Phys.* **14**, 451–455 (2018).
71. Ikhlas, M. et al. Large anomalous Nernst effect at room temperature in a chiral antiferromagnet. *Nat. Phys.* **13**, 1085–1090 (2017).
72. Suzuki, T. et al. Large anomalous Hall effect in a half-Heusler antiferromagnet. *Nat. Phys.* **12**, 1119–1123 (2016).
73. Liang, T. et al. Anomalous Nernst effect in the Dirac semimetal Cd_3As_2 . *Phys. Rev. Lett.* **118**, 136601 (2017).
74. Morimoto, T. & Nagaosa, N. Topological nature of nonlinear optical effects in solids. *Sci. Adv.* **2**, e1501524 (2016).
75. Wu, L. et al. Giant anisotropic nonlinear optical response in transition metal monophosphide Weyl semimetals. *Nat. Phys.* **13**, 350–355 (2017).
76. Chen, R. Y. et al. Magneto-infrared spectroscopy of Landau levels and Zeeman splitting of three-dimensional massless Dirac fermions in ZrTe_5 . *Phys. Rev. Lett.* **115**, 176404 (2015).
77. Akrap, A. et al. Magneto-optical signature of massless Kane electrons in Cd_3As_2 . *Phys. Rev. Lett.* **117**, 136401 (2016).
78. Polatkan, S. et al. Magneto-optics of a Weyl semimetal beyond the conical band approximation: case study of TaP. *Phys. Rev. Lett.* **124**, 176402 (2020).
79. Okamura, Y. et al. Giant magneto-optical responses in magnetic Weyl semimetal $\text{Co}_3\text{Sn}_2\text{S}_2$. *Nat. Commun.* **11**, 4619 (2020).
80. Jiang, Y. et al. Landau quantization in coupled Weyl points: a case study of semimetal NbP. *Nano Lett.* **18**, 7726–7731 (2018).
81. Yuan, X. et al. Chiral Landau levels in Weyl semimetal NbAs with multiple topological carriers. *Nat. Commun.* **9**, 1854 (2018).
82. Son, D. T. & Yamamoto, N. Berry curvature, triangle anomalies, and the chiral magnetic effect in Fermi liquids. *Phys. Rev. Lett.* **109**, 181602 (2012).
83. Grushin, A. G. Consequences of a condensed matter realization of Lorentz-violating QED in Weyl semimetals. *Phys. Rev. D* **86**, 045001 (2012).
84. Song, Z. & Dai, X. Hear the sound of Weyl fermions. *Phys. Rev. X* **9**, 021053 (2019).
85. Xiang, J. et al. Giant magnetic quantum oscillations in the thermal conductivity of TaAs: indications of chiral zero sound. *Phys. Rev. X* **9**, 031036 (2019).
86. Zhang, Y. et al. Photogalvanic effect in Weyl semimetals from first principles. *Phys. Rev. B* **97**, 241118 (2018).
87. Ma, Q. et al. Direct optical detection of Weyl fermion chirality in a topological semimetal. *Nat. Phys.* **13**, 842–847 (2017).
88. Chan, C.-K., Lindner, N. H., Refael, G. & Lee, P. A. Photocurrents in Weyl semimetals. *Phys. Rev. B* **95**, 041104 (2017).
89. Kharzeev, D. E., Kikuchi, Y., Meyer, R. & Tanizaki, Y. Giant photocurrent in asymmetric Weyl semimetals from the helical magnetic effect. *Phys. Rev. B* **98**, 014305 (2018).
90. Wang, Q. et al. Robust edge photocurrent response on layered type II Weyl semimetal WTe_2 . *Nat. Commun.* **10**, 5736 (2019).
91. Rees, D. et al. Helicity-dependent photocurrents in the chiral Weyl semimetal RhSi. *Sci. Adv.* **6**, eaba0509 (2020).
92. Ji, Z. et al. Spatially dispersive circular photogalvanic effect in a Weyl semimetal. *Nat. Mater.* **18**, 955–962 (2019).
93. Ma, J. et al. Nonlinear photoresponse of type-II Weyl semimetals. *Nat. Mater.* **18**, 476–481 (2019).
94. Osterhoudt, G. B. et al. Colossal mid-infrared bulk photovoltaic effect in a type-I Weyl semimetal. *Nat. Mater.* **18**, 471–475 (2019).
95. Kaushik, S., Kharzeev, D. E. & Philip, E. J. Chiral magnetic photocurrent in Dirac and Weyl materials. *Phys. Rev. B* **99**, 075150 (2019).
96. de Juan, F., Grushin, A. G., Morimoto, T. & Moore, J. E. Quantized circular photogalvanic effect in Weyl semimetals. *Nat. Commun.* **8**, 15995 (2017).
97. Sodemann, I. & Fu, L. Quantum nonlinear Hall effect induced by Berry curvature dipole in time-reversal invariant materials. *Phys. Rev. Lett.* **115**, 216806 (2015).
98. Zhang, Y., Sun, Y. & Yan, B. Berry curvature dipole in Weyl semimetal materials: an ab initio study. *Phys. Rev. B* **97**, 041101 (2018).
99. Vafeek, O. & Vishwanath, A. Dirac fermions in solids: from high-Tc cuprates and graphene to topological insulators and Weyl semimetals. *Annu. Rev. Condens. Matter Phys.* **5**, 83–112 (2014).
100. Burkov, A. A. Chiral anomaly and transport in Weyl metals. *J. Phys. Condens. Matter* **27**, 113201 (2015).
101. Hosur, P. & Qi, X. Recent developments in transport phenomena in Weyl semimetals. *C. R. Phys.* **14**, 857–870 (2013).
102. Yan, B. & Felser, C. Topological materials: Weyl semimetals. *Annu. Rev. Condens. Matter Phys.* **8**, 337–354 (2017).
103. Burkov, A. A. Weyl metals. *Annu. Rev. Condens. Matter Phys.* **9**, 359–378 (2018).
104. Hasan, M. Z., Xu, S.-Y., Belopolski, I. & Huang, S.-M. Discovery of Weyl fermion semimetals and topological Fermi arc states. *Annu. Rev. Condens. Matter Phys.* **8**, 289–309 (2017).
105. Jia, S., Xu, S.-Y. & Hasan, M. Z. Weyl semimetals, Fermi arcs and chiral anomalies. *Nat. Mater.* **15**, 1140–1144 (2016).
106. Burkov, A. A. Topological semimetals. *Nat. Mater.* **15**, 1145–1148 (2016).
107. Fang, C., Weng, H., Dai, X. & Fang, Z. Topological nodal line semimetals. *Chin. Phys. B* **25**, 117106 (2016).
108. Wang, A.-Q., Ye, X.-G., Yu, D.-P. & Liao, Z.-M. Topological semimetal nanostructures: from properties to topotronics. *ACS Nano* **14**, 3755–3778 (2020).
109. Zhang, C., Lu, H.-Z., Shen, S.-Q., Chen, Y. P. & Xiu, F. Towards the manipulation of topological states of matter: a perspective from electron transport. *Sci. Bull.* **63**, 580–594 (2018).
110. Lu, H.-Z. & Shen, S.-Q. Quantum transport in topological semimetals under magnetic fields. *Front. Phys.* **12**, 127201 (2017).
111. Sun, H.-P. & Lu, H.-Z. Quantum transport in topological semimetals under magnetic fields (II). *Front. Phys.* **14**, 334005 (2019).
112. Gao, H., Venderbos, J. W. F., Kim, Y. & Rappe, A. M. Topological semimetals from first principles. *Annu. Rev. Mater. Res.* **49**, 153–183 (2019).
113. Hu, J., Xu, S.-Y., Ni, N. & Mao, Z. Transport of topological semimetals. *Annu. Rev. Mater. Res.* **49**, 207–252 (2019).
114. Bernevig, A., Weng, H., Fang, Z. & Dai, X. Recent progress in the study of topological semimetals. *J. Phys. Soc. Jpn.* **87**, 041001 (2018).
115. Wang, S., Lin, B.-C., Wang, A.-Q., Yu, D.-P. & Liao, Z.-M. Quantum transport in Dirac and Weyl semimetals: a review. *Adv. Phys. X* **2**, 518–544 (2017).
116. Schoop, L. M., Pielnhofer, F. & Lotsch, B. V. Chemical principles of topological semimetals. *Chem. Mater.* **30**, 3155–3176 (2018).
117. Zhang, Y. Cyclotron orbit knot and tunable-field quantum Hall effect. *Phys. Rev. Res.* **1**, 022005 (2019).
118. Peri, V., Dubček, T., Valenti, A., Ilan, R. & Huber, S. D. Weyl orbits without an external magnetic field. *Phys. Rev. B* **101**, 235117 (2020).
119. Kourtis, S., Li, J., Wang, Z., Yazdani, A. & Bernevig, B. A. Universal signatures of Fermi arcs in quasiparticle interference on the surface of Weyl semimetals. *Phys. Rev. B* **93**, 041109 (2016).
120. Batabyal, R. et al. Visualizing weakly bound surface Fermi arcs and their correspondence to bulk Weyl fermions. *Sci. Adv.* **2**, e1600709 (2016).
121. Zheng, H. et al. Atomic-scale visualization of quantum interference on a Weyl semimetal surface by scanning tunneling microscopy. *ACS Nano* **10**, 1378–1385 (2016).
122. Inoue, H. et al. Quasiparticle interference of the Fermi arcs and surface-bulk connectivity of a Weyl semimetal. *Science* **351**, 1184–1187 (2016).
123. Analytis, J. G. et al. Two-dimensional surface state in the quantum limit of a topological insulator. *Nat. Phys.* **6**, 960–964 (2010).
124. Peng, H. et al. Aharonov–Bohm interference in topological insulator nanoribbons. *Nat. Mater.* **9**, 225–229 (2010).
125. Kong, D. et al. Ambipolar field effect in the ternary topological insulator $(\text{Bi}_x\text{Sb}_{1-x})_2\text{Te}_3$ by composition tuning. *Nat. Nanotechnol.* **6**, 705–709 (2011).
126. Zhang, J. et al. Band structure engineering in $(\text{Bi}_{1-x}\text{Sb}_x)_2\text{Te}_3$ ternary topological insulators. *Nat. Commun.* **2**, 574 (2011).
127. Xiu, F. et al. Manipulating surface states in topological insulator nanoribbons. *Nat. Nanotechnol.* **6**, 216–221 (2011).
128. Moll, P. J. W. et al. Transport evidence for Fermi-arc-mediated chirality transfer in the Dirac semimetal Cd_3As_2 . *Nature* **535**, 266–270 (2016).
A transport study of the Weyl orbit.
129. Zhang, C. et al. Evolution of Weyl orbit and quantum Hall effect in Dirac semimetal Cd_3As_2 . *Nat. Commun.* **8**, 1272 (2017).
Report of quantum Hall effect in Cd_3As_2 .
130. Zheng, G. et al. Recognition of Fermi-arc states through the magnetoresistance quantum oscillations in Dirac semimetal Cd_3As_2 nanoplates. *Phys. Rev. B* **96**, 121407 (2017).
131. Zhang, C. et al. Ultrahigh conductivity in Weyl semimetal NbAs nanobelts. *Nat. Mater.* **18**, 482–488 (2019).
This work shows that the Fermi arc structure can suppress electron scattering and achieve ultrahigh conductivity.
132. Zhang, C. et al. Quantum Hall effect based on Weyl orbits in Cd_3As_2 . *Nature* **565**, 331–336 (2019).
This paper presents evidences of 3D quantum Hall effect based on Weyl orbits.
133. Schumann, T. et al. Observation of the quantum Hall effect in confined films of the three-dimensional Dirac semimetal Cd_3As_2 . *Phys. Rev. Lett.* **120**, 016801 (2018).
134. Li, P. et al. Evidence for topological type-II Weyl semimetal WTe_2 . *Nat. Commun.* **8**, 2150 (2017).
135. Nair, N. L. et al. Signatures of possible surface states in TaAs. *Phys. Rev. B* **102**, 075402 (2020).
136. Resta, G., Pi, S.-T., Wan, X. & Savrasov, S. Y. High surface conductivity of Fermi-arc electrons in Weyl semimetals. *Phys. Rev. B* **97**, 085142 (2018).
137. Wang, L.-X., Li, C.-Z., Yu, D.-P. & Liao, Z.-M. Aharonov–Bohm oscillations in Dirac semimetal Cd_3As_2 nanowires. *Nat. Commun.* **7**, 10769 (2016).
138. Lin, B.-C. et al. Gate-tuned Aharonov–Bohm interference of surface states in a quasiballistic Dirac semimetal nanowire. *Phys. Rev. B* **95**, 235436 (2017).
139. Jauregui, L. A., Pettes, M. T., Rokhsinon, L. P., Shi, L. & Chen, Y. P. Magnetic field-induced helical mode and topological transitions in a topological insulator nanoribbon. *Nat. Nanotechnol.* **11**, 345–351 (2016).
140. Halperin, B. I. Possible states for a three-dimensional electron gas in a strong magnetic field. *Jpn. J. Appl. Phys.* **26**, 1913 (1987).
141. Koshino, M., Aoki, H., Kuroki, K., Kagoshima, S. & Osada, T. Hofstadter butterfly and integer quantum Hall effect in three dimensions. *Phys. Rev. Lett.* **86**, 1062–1065 (2001).
142. Bernevig, B. A., Hughes, T. L., Raghu, S. & Arovas, D. P. Theory of the three-dimensional quantum Hall effect in graphite. *Phys. Rev. Lett.* **99**, 146804 (2007).
143. Uchida, M. et al. Quantum Hall states observed in thin films of Dirac semimetal Cd_3As_2 . *Nat. Commun.* **8**, 2274 (2017).
144. Nishihaya, S. et al. Gate-tuned quantum Hall states in Dirac semimetal $(\text{Cd}_{1-x}\text{Zn}_x)_3\text{As}_2$. *Sci. Adv.* **4**, eaar5668 (2018).
145. Nishihaya, S. et al. Quantized surface transport in topological Dirac semimetal films. *Nat. Commun.* **10**, 2564 (2019).
This paper shows the distinct quantum Hall origins in Cd_3As_2 films at different thickness regimes.
146. Galletti, L. et al. Two-dimensional Dirac fermions in thin films of Cd_3As_2 . *Phys. Rev. B* **97**, 115132 (2018).
147. Goyal, M. et al. Surface states of strained thin films of the Dirac semimetal Cd_3As_2 . *Phys. Rev. Mater.* **3**, 064204 (2019).
148. Lin, B.-C. et al. Observation of an odd-integer quantum Hall effect from topological surface states in Cd_3As_2 . *Phys. Rev. Lett.* **122**, 036602 (2019).
149. Nishihaya, S. et al. Intrinsic coupling between spatially-separated surface Fermi-arcs in Weyl orbit quantum Hall states. *Nat. Commun.* **12**, 2572 (2021).
This paper shows the coupling between Fermi arcs of opposite surfaces in the quantum Hall state of Cd_3As_2 .
150. Xu, Y., Miotkowski, I. & Chen, Y. P. Quantum transport of two-species Dirac fermions in dual-gated three-dimensional topological insulators. *Nat. Commun.* **7**, 11434 (2016).
151. Kealhofer, D. A. et al. Basal-plane growth of cadmium arsenide by molecular beam epitaxy. *Phys. Rev. Mater.* **3**, 031201 (2019).

152. Kealhofer, D. A., Galletti, L., Schumann, T., Suslov, A. & Stemmer, S. Topological insulator state and collapse of the quantum Hall effect in a three-dimensional Dirac semimetal heterojunction. *Phys. Rev. X* **10**, 011050 (2020).
153. Kargarian, M., Randeria, M. & Lu, Y.-M. Are the surface Fermi arcs in Dirac semimetals topologically protected? *Proc. Natl Acad. Sci. USA* **113**, 8648 (2016).
154. Arribi, P. V. et al. Topological surface states in strained Dirac semimetal thin films. *Phys. Rev. B* **102**, 155141 (2020).
155. Baum, Y., Berg, E., Parameswaran, S. A. & Stern, A. Current at a distance and resonant transparency in Weyl semimetals. *Phys. Rev. X* **5**, 041046 (2015).
156. Hou, Z. & Sun, Q.-F. Nonlocal correlation mediated by Weyl orbits. *Phys. Rev. Res.* **2**, 023236 (2020).
157. Bulmash, D. & Qi, X.-L. Quantum oscillations in Weyl and Dirac semimetal ultrathin films. *Phys. Rev. B* **93**, 081103 (2016).
158. Ominato, Y. & Koshino, M. Magnetotransport in Weyl semimetals in the quantum limit: role of topological surface states. *Phys. Rev. B* **93**, 245304 (2016).
159. Liu, T., Pikulin, D. I. & Franz, M. Quantum oscillations without magnetic field. *Phys. Rev. B* **95**, 041201 (2017).
160. Behrends, J., Roy, S., Kolodrubetz, M. H., Bardarson, J. H. & Grushin, A. G. Landau levels, Bardeen polynomials, and Fermi arcs in Weyl semimetals: lattice-based approach to the chiral anomaly. *Phys. Rev. B* **99**, 140201 (2019).
161. Ilan, R., Grushin, A. G. & Pikulin, D. I. Pseudo-electromagnetic fields in 3D topological semimetals. *Nat. Rev. Phys.* **2**, 29–41 (2020).
162. Bovenzi, N., Breitzkreiz, M., O'Brien, T. E., Tworzydło, J. & Beenakker, C. W. J. Twisted Fermi surface of a thin-film Weyl semimetal. *New J. Phys.* **20**, 023023 (2018).
163. Yang, G. & Zhang, Y. Quantum-limit Hall effect with large carrier density in topological semimetals. *Phys. Rev. B* **103**, L241104 (2021).
164. Wang, H. et al. Observation of superconductivity induced by a point contact on 3D Dirac semimetal Cd_3As_2 crystals. *Nat. Mater.* **15**, 38–42 (2016).
165. Aggarwal, L. et al. Unconventional superconductivity at mesoscopic point contacts on the 3D Dirac semimetal Cd_3As_2 . *Nat. Mater.* **15**, 32–37 (2016).
166. Bachmann, M. D. et al. Inducing superconductivity in Weyl semimetal microstructures by selective ion sputtering. *Sci. Adv.* **3**, e1602983 (2017).
167. Huang, C. et al. Proximity-induced surface superconductivity in Dirac semimetal Cd_3As_2 . *Nat. Commun.* **10**, 2217 (2019).
168. Huang, C. et al. Inducing strong superconductivity in WTe_2 by a proximity effect. *ACS Nano* **12**, 7185–7196 (2018).
169. Pan, X.-C. et al. Pressure-driven dome-shaped superconductivity and electronic structural evolution in tungsten ditelluride. *Nat. Commun.* **6**, 7805 (2015).
170. Wang, A.-Q. et al. 4π -Periodic supercurrent from surface states in Cd_3As_2 nanowire-based Josephson junctions. *Phys. Rev. Lett.* **121**, 237701 (2018).
171. Wang, S., Lin, B.-C., Zheng, W.-Z., Yu, D. & Liao, Z.-M. Fano interference between bulk and surface states of a Dirac semimetal Cd_3As_2 nanowire. *Phys. Rev. Lett.* **120**, 257701 (2018).
172. Li, C.-Z. et al. Reducing electronic transport dimension to topological hinge states by increasing geometry size of Dirac semimetal Josephson junctions. *Phys. Rev. Lett.* **124**, 156601 (2020).
173. Li, C.-Z. et al. Fermi-arc supercurrent oscillations in Dirac semimetal Josephson junctions. *Nat. Commun.* **11**, 1150 (2020).
174. He, L. et al. Pressure-induced superconductivity in the three-dimensional topological Dirac semimetal Cd_3As_2 . *NPJ Quant. Mater.* **1**, 16014 (2016).
175. Aggarwal, L. et al. Mesoscopic superconductivity and high spin polarization coexisting at metallic point contacts on Weyl semimetal TaAs . *Nat. Commun.* **8**, 13974 (2017).
176. Wang, H. et al. Discovery of tip induced unconventional superconductivity on Weyl semimetal. *Sci. Bull.* **62**, 425–430 (2017).
177. Qi, Y. et al. Superconductivity in Weyl semimetal candidate MoTe_2 . *Nat. Commun.* **7**, 11038 (2016).
178. Wang, W. et al. Evidence for an edge supercurrent in the Weyl superconductor MoTe_2 . *Science* **368**, 534–537 (2020).
179. Song, J. C. W. & Rudner, M. S. Fermi arc plasmons in Weyl semimetals. *Phys. Rev. B* **96**, 205443 (2017).
180. Rajamathi, C. R. et al. Weyl semimetals as hydrogen evolution catalysts. *Adv. Mater.* **29**, 1606202 (2017).
181. Li, G. et al. Carbon-tailored semimetal MoP as an efficient hydrogen evolution electrocatalyst in both alkaline and acid media. *Adv. Energy Mater.* **8**, 1801258 (2018).
182. Li, G. et al. Surface states in bulk single crystal of topological semimetal $\text{Co}_2\text{Sn}_2\text{S}_6$ toward water oxidation. *Sci. Adv.* **5**, eaaw9867 (2019).
183. Li, J. et al. Topological quantum catalyst: Dirac nodal line states and a potential electrocatalyst of hydrogen evolution in the TiSi family. *Sci. China Mater.* **61**, 23–29 (2018).
184. He, Yuan. et al. Topologically nontrivial $1\text{T}'\text{-MoTe}_2$ as highly efficient hydrogen evolution electrocatalyst. *J. Phys. Mater.* **4**, 014001 (2021).
185. Zhao, G.-Q., Rui, W. B., Wang, C. M., Lu, H.-Z. & Xie, X. C. Quantized Hall conductance in 3D topological nodal-line semimetals without chiral symmetry. Preprint at <https://arxiv.org/abs/2004.01386> (2020).
186. Shi, L. & Song, J. C. Large optical conductivity of Dirac semimetal Fermi arc surface states. *Phys. Rev. B* **96**, 081410 (2017).
187. Chang, G. et al. Unconventional photocurrents from surface Fermi arcs in topological chiral semimetals. *Phys. Rev. Lett.* **124**, 166404 (2020).

Acknowledgements

F.X. was supported by the National Natural Science Foundation of China (grant nos. 11934005 and 11874116), National Key Research and Development Program of China (grant nos. 2017YFA0303302 and 2018YFA0305601), the Science and Technology Commission of Shanghai (grant no. 19511120500), the Shanghai Municipal Science and Technology Major Project (grant no. 2019SHZDZX01) and the Program of Shanghai Academic/Technology Research Leader (grant no. 20XD1400200). C.Z. was sponsored by Shanghai Sailing Program (grant no. 20YF1402300), Natural Science Foundation of Shanghai (grant no. 20ZR1407500) and the start-up grant at Fudan University. Y.Z. was supported by the start-up grant at Peking University.

Author contributions

C.Z. and F.X. drafted the manuscript and all the authors contributed to finalizing the manuscript.

Competing interests

The authors declare no competing interests.

Peer review information

Nature Reviews Physics thanks Hongming Weng and Yan Sun for their contribution to the peer review of this work.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

© Springer Nature Limited 2021, corrected publication 2021